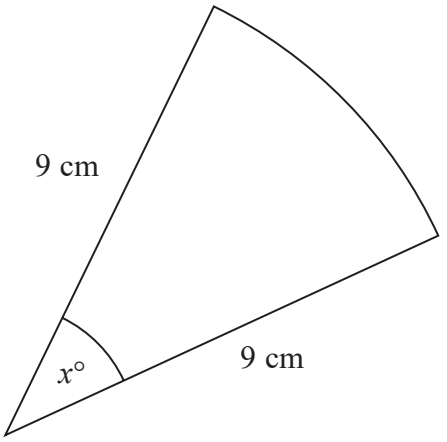
19 The diagram shows a sector of a circle of radius 9 cm.



The sector has a perimeter of 25 cm.

Work out the value of x.
Give your answer correct to 1 decimal place.

Arc length = 25 – 9 – 9 = 7cm.

1 mark
Remove both radii

$$\frac{7}{18\pi} = \frac{x}{360}$$

1 mark
Same circle fraction

$x = 7 \times 360 \div (18\pi) = 44.563384...°$.

1 mark
Solve for angle

$x = 44.6°$ (1 decimal place).

1 mark
44.6°

(Total for Question 19 is 4 marks)